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AI and cloud are changing virtualization

AI is “another tool set that just complicates the mash-up when it comes to containerized stacks sitting on top of other operating technologies,” NCC expert says.

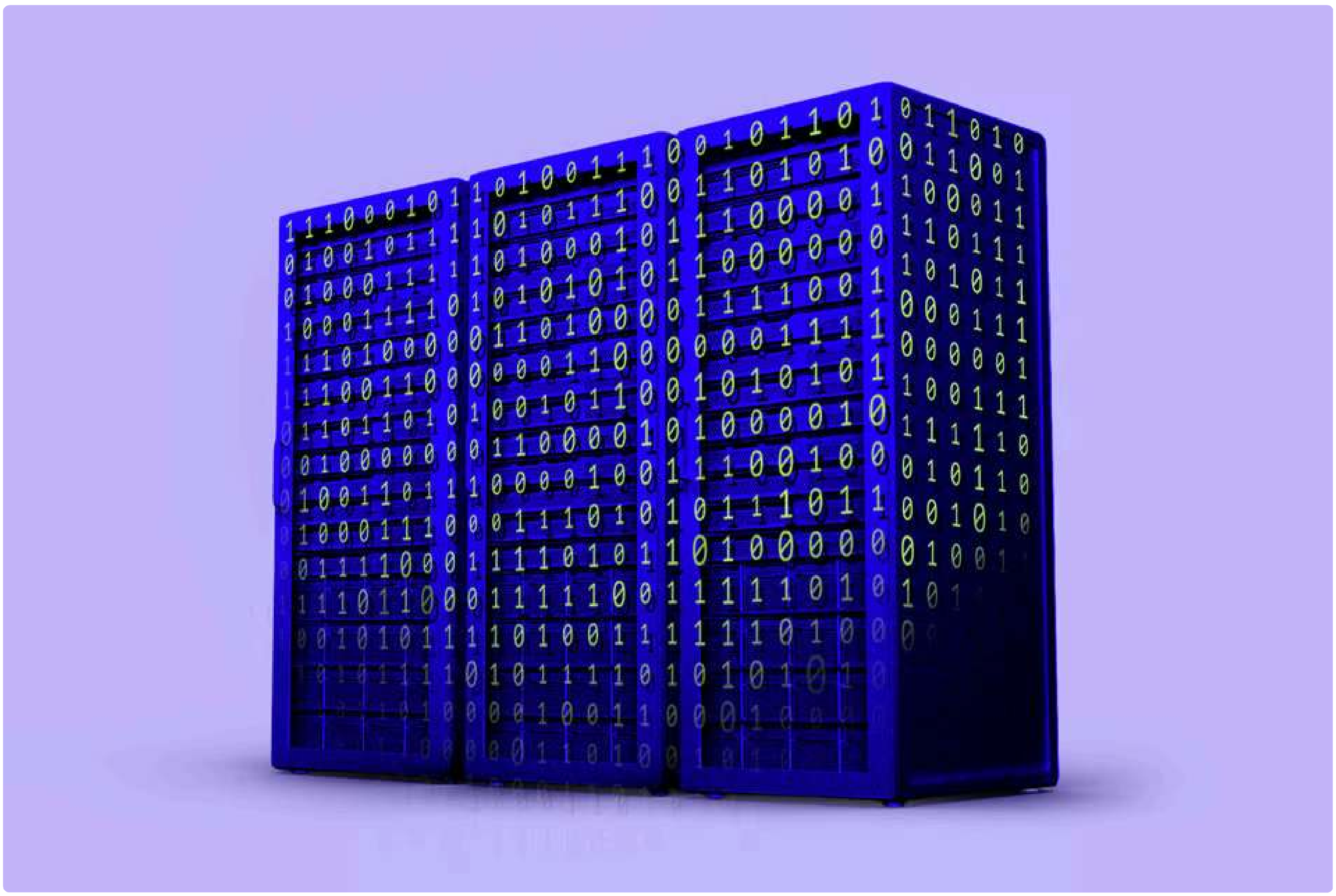

Illustration: Amelia Kinsinger, Photos: Adobe Stock

By Eoin Higgins

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Changing times, changing priorities—the impact of AI on modern virtualization has made things very different for IT pros.

For over two decades, virtualization has been an IT workhorse, allowing IT pros to expand their tech stacks' storage and capabilities. The maturing of the cloud and the introduction of AI have made virtual environments even more complex, even if virtualization itself, in the words of Forrester analyst Devin Dickerson, is "not in a dissimilar state to what it's been in the last couple years."

"There's the potential for disruption to the space because AI features, over time, will become expected—so things like predictive and self-healing infrastructure, and something that can understand usage patterns, optimize right size in real time, anticipate demand, dealing with performance, rerouting workloads, without necessarily having human intervention," Dickerson said. "Those can become expectations over time, the more that they're baked into managed platforms."

Reality bites

Part of the problem with virtualization is the perception of the technology as legacy rather than modern architecture. Dickerson said he believes that's due at least in part to the use of containerization and microservices, which have existed for many years, and not to the potential of AI and cloud—but that the latter can be seen as a variation on a theme rather than a brand new tune.

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"AI workloads actually bring to bear a part of cloud computing that used to be really hyped and popular, but also is less visible to enterprise decision-makers, which is serverless in the cloud," Dickerson said.

That may open the door to a whole new set of opportunities. In order to manage the escalating demands of AI on virtualization, ConductorOne CISO Kevin Paige told IT Brew, we need to treat the integrated technology as a necessary architecture in the IT ecosystem. Virtualization is coupled with other technologies that overlap heavily with AI like containerization, GPU partitioning, and microservices; security is, as always, an overarching concern.

"I'm seeing virtualization fall into all these categories as an architectural principle, to help things be a little bit more secure, but also to scale very quickly to meet the needs of AI, whether it's on the cloud or on-prem," Paige said.

Getting bigger

AI could help virtualization scale up management of containerization and other microservices within virtual environments. It may also expand the threat surface.

As NCC Group Director and Senior Adviser Nigel Gibbons told IT Brew, AI is "another tool set that just complicates the mash-up when it comes to containerized stacks sitting on top of other operating technologies."

"It can magnify that risk surface, because if those core, foundational operational and platform control plane imperatives are not already in place, then it's just accelerating," Gibbons, whose focus is global cloud security services, said.

> *"That's what AI does, because it doesn't have those native control boundaries that are designed in so that risk is managed through policy, through code, identity*

> *containment, data flow containment—all these core disciplines that you would expect off a sort of a platform control plane are absent.”*

For security-minded IT pros, the reality is that the only way to ever truly protect yourself is just turn the entire system off. That’s unlikely, so making sure that there are layers of separation—that your permissions are segmented as needed—is a good first step.

“These layers of abstraction allow us to take unique security capabilities to reduce the blast radius, turn things on when they’re necessary, turn them off when they’re not,” Paige said. “We can do that from an access perspective.”

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ABOUT THE AUTHOR

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Eoin Higgins is a reporter for IT Brew whose work focuses on the AI sector and IT operations and strategy.

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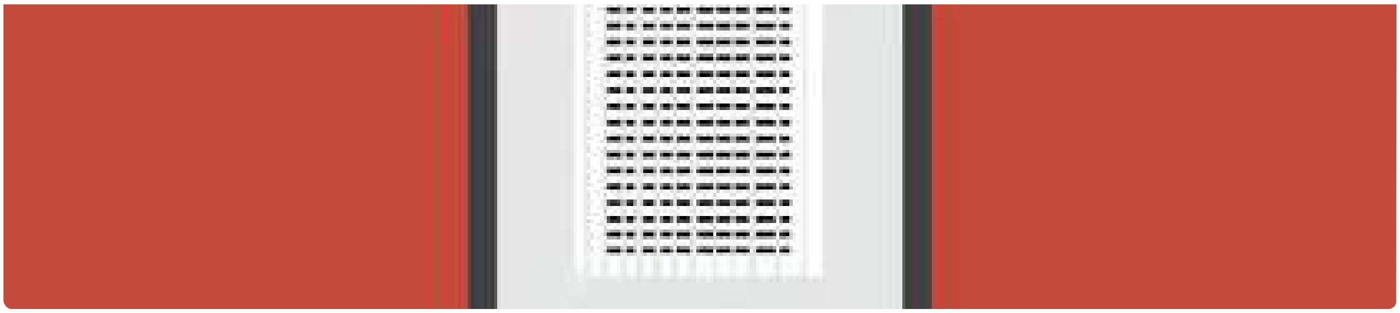

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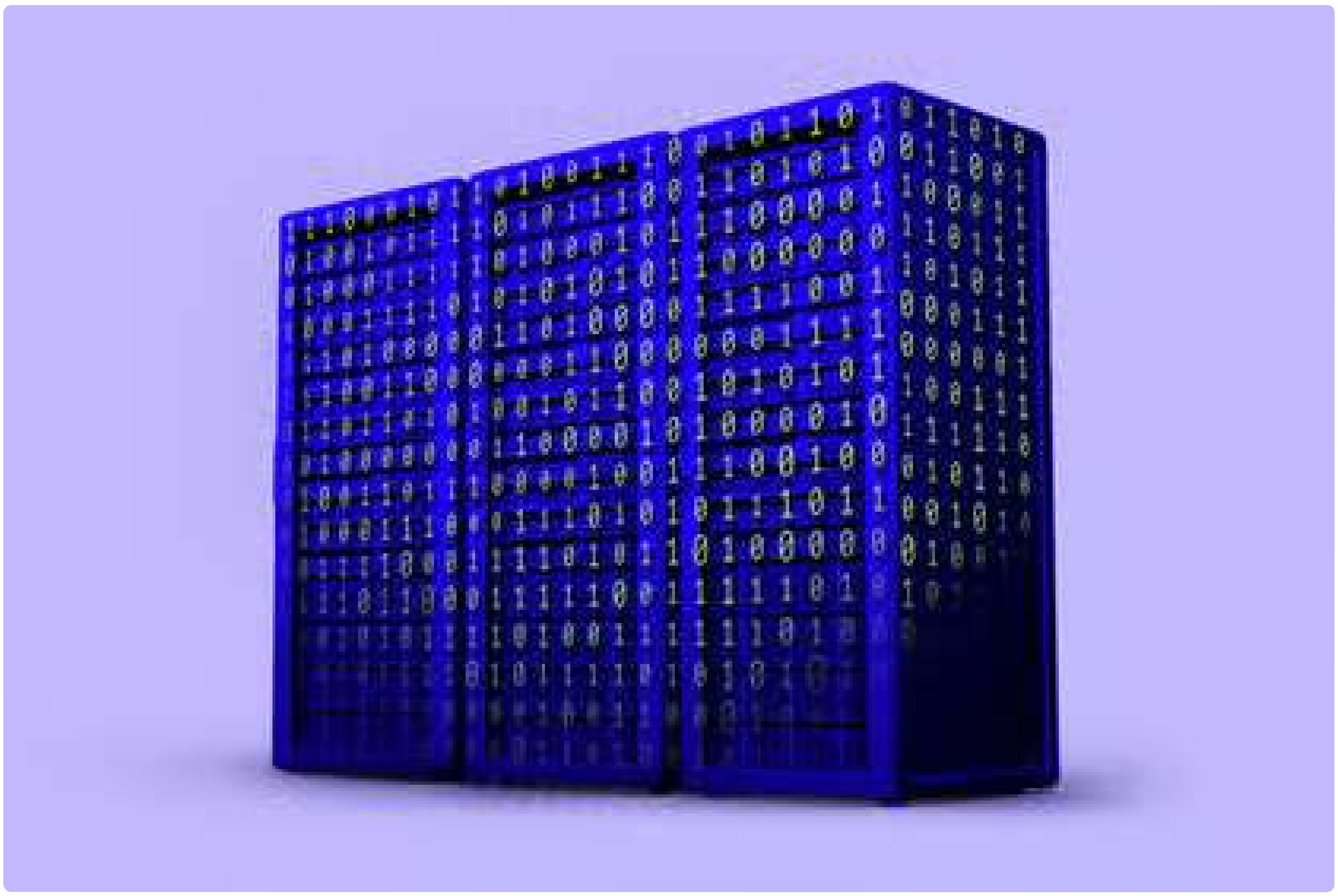

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